Dynamic Performance Characteristics of a Porous Volumetric Solar Receiver Under Transient Flux Conditions

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The solar flux incident on a volumetric receiver is inherently unsteady, resulting in high thermal stresses, fatigue failures, and reduced component life. The knowledge of transient response characteristics of a porous volumetric receiver used in concentrating solar technologies is cardinal for its reliable and safe working. The dynamic controlling of the solar-to-thermal conversion process is also possible with the prior prediction of the output variations. The present study aims to investigate the transient behavior of a porous volumetric receiver subjected to flux variations approximations occurring in real working scenarios with the help of a coupled transient model. The solid and fluid temperature fields, output fluid temperature, and pressure drop variations are determined for transient flux conditions during start-up, shut-down, clear sky, and cloud passage. The results are used to analyze the thermal response of the receiver during various operating conditions. In addition, the effects of structural parameters of the porous absorber are also investigated. The results indicate that the receiver transient performance is comparatively more affected by the variation in porosity than in pore size for all conditions. Smaller porosities and pore sizes show slower thermal response to transient fluctuations and less temperature changes during cloud passage. Conversely, higher values help in the faster restoration of the steady-state output conditions without dynamic control. [DOI: 10.1115/1.4056622]

Keywords: volumetric receiver, porous media, coupled transient simulation, heat transfer, thermal radiation

1 Introduction

The rapid development of the global economy has increased energy demands remarkably. Efficiency improvements in energy production and replacing fossil fuels with renewable energy sources are the typical strategies for sustainable energy development [1]. The abundant and free availability of solar energy and growing output efficiencies make it the best option for the future world [2]. Concentrated solar power (CSP) plants have the potential to improve the flexibility of power systems by ensuring balanced and reliable power generation [3]. The solar power tower is the most promising due to its higher theoretical maximum efficiency and better electrical block efficiency [4]. It consists of a heliostat field, receiver, block for power production, and thermal storage system [5]. A detailed review of various central receiver designs employed for high temperature power cycles was given by Ho and Iverson [6]. The volumetric receivers consist of absorbers made of porous foams or honeycomb structures [7]. More focus has been given to volumetric receivers due to their flexibility in functionality with cheaper, simpler, and more efficient design than tubular receivers [8]. Volumetric receivers with honeycomb structures have been studied in detail by [9–12].

A considerable amount of work has been done on the steady-state performance of the porous volumetric receivers. Xu et al. [13] modeled a porous media subjected to a uniform solar flux included as a boundary condition and compared the results for various correlations of volumetric heat transfer coefficients available in the literature. Wu et al. [14] developed a coupled numerical model for fluid flow and heat transfer. The complete 3D numerical studies for a porous volumetric absorber were performed by Baretto et al. [15,16]. The optical modeling for the concentrator-receiver system and solar radiation absorption within the absorber were computed using Monte Carlo ray tracing (MCRT). Studies have shown the potential of graded porous media in enhancing the performance of porous receivers. Roldán et al. [17] simulated gradual porosity receiver configurations in radial and axial directions with porosity variations. The results for axial variation showed better performance in terms of efficiency. A combination of experimental and numerical work for graded porous media was done by Du et al. [18]. An optimization study using a genetic algorithm was also included to find the optimum porosity distribution for best performance.

In actual working conditions, a steady-state seldom exists. The receiver is exposed to cyclic working due to diurnal cycles and intermittent incident fluxes with interruptions caused by clouds [19]. Such fluctuations may result in premature failure of the receiver material due to excessive thermal stresses and fatigue damages [20]. A few authors have investigated the transient performance of solar receivers. Xu et al. [21] numerically studied the unsteady behavior of an external solar receiver. The effect of initial conditions, direct normal irradiance (DNI) level, tube thickness, and wind speed on the receiver's performance was analyzed. Soo Too et al. developed a transient optical-thermal model with a closed loop aiming strategy for molten salt solar receivers [20]. Concerning porous volumetric solar receivers, a transient thermal model was implemented by Ren et al. [22]. The effect of radiation heat transfer was considered by adding a radiative conductivity term in solid thermal conductivity. The transient response of the receiver was investigated under sudden supply and loss of heat flux. A

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similar study was performed by Reddy and Nataraj [23], and porosity's effect on thermal response was investigated for flux supply and cut. Wu and Wang [24] developed a fully coupled transient model and compared the results with experimental data. The effect of sudden supply, loss, and step changes in the heat flux was investigated. A study to investigate the effects of thermophysical properties of fluid and solid phase of the porous volumetric receiver on the transient thermal response was done by Fuqiang et al. [25]. The MCRT was implemented to obtain the heat flux distribution at the front of the receiver. At the same time, the P1 approximation was used to model radiation within the receiver. Ribeiro and de Lemos [26] developed an in-house code to analyze the transient behavior and thermal efficiency of porous volumetric receivers. The radiation within the receiver was modeled using the Rosseland approximation while the transient flux was applied as a boundary condition at the inlet. A parametric study by varying permeability, porosity, inlet velocity, and thermal conductivity ratio was carried out to investigate their effects on transient performance. Zoller et al. [27] developed a transient heat transfer model for porous solar receiver-reactor for thermochemical redox cycling. A few researchers have coupled the transient response of receivers with dynamic controlling to keep the fluctuations within limits [20,28].

The literature review suggests that the transient performance of porous volumetric receivers is less explored as compared to steady-state performance. Thus, this work is motivated by the lack of systematic transient studies in the literature as well as an absence of a clear connection between flux variations considered in previous studies and real working conditions. In the present work, the transient thermal response of a porous volumetric receiver during the operation cycle of a CSP is investigated while incorporating incident flux fluctuations close to real scenarios. A detailed analysis is carried out to extend the understanding of the transient behavior of the receiver during the start-up and shut-down period. In addition, the transient response of the volumetric receiver is studied during a clear sky day and cloud passing. The effects of porosity and pore size on the time-dependent temperature fields during various working scenarios are also investigated.

2 Model Description

A porous cylindrical absorber of 5 cm in length and 2.5 cm in radius is selected for the present study. A homogeneous equivalent method is implemented to model the porous absorber. Using this method, the porous medium is considered as a homogeneous effective medium consisting of two continuous phases, solid and fluid [29]. Silicon carbide (SiC) is selected as the material of the porous absorber due to its superior thermal properties, including enhanced absorptance and high thermal conductivity [30]. The air is selected as the heat transfer fluid flowing through the absorber. Figure 1 presents the schematic of the computational domain.

2.1 Fluid Flow Model. The flow field is considered transient, compressible, and laminar. The macroscopic continuity and momentum transport equations are solved for air flow inside a

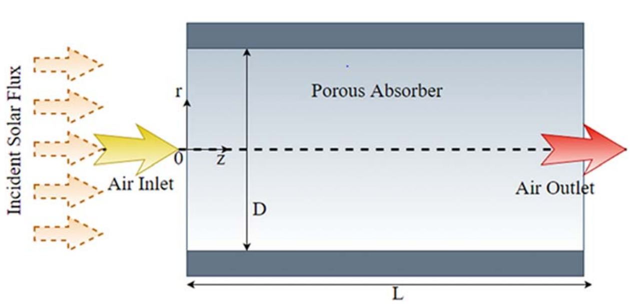

Fig. 1 Schematic of the computational domain

porous medium.

$$\frac{\partial}{\partial t}(\phi\rho_f)+\nabla\cdot(\rho_f\boldsymbol{U})=0 \quad (1)$$

$$\frac{1}{\phi}\rho_f\frac{\partial \boldsymbol{U}}{\partial t}+\frac{1}{\phi}(\rho_f\boldsymbol{U}\cdot\nabla)\boldsymbol{U}\frac{1}{\phi}=-\nabla p+\nabla\cdot\left[\frac{\mu}{\phi}\left(\nabla\boldsymbol{U}+(\nabla\boldsymbol{U})'-\frac{2}{3}\nabla\cdot\boldsymbol{U}I\right)\right]-\left(\frac{\mu}{K}+\frac{C_F}{\sqrt{K}}\rho_f|\boldsymbol{U}|\right)\boldsymbol{U} \quad (2)$$

$\boldsymbol{U}=\phi\boldsymbol{u}$ denotes the superficial velocity where ϕ is the porosity and u is the pore velocity. The second and third terms on the right-hand side of Eq. (2) are the Brinkman and Forchheimer terms, respectively. While the Brinkman term accounts for the shear stresses, the Forchheimer term adds a viscous drag force [31]. $K=\frac{d_p^2}{1039-1002\phi}$ and $C_F=\frac{0.5138\phi^{-5.739}}{d_p}\sqrt{K}$ denote the permeability and inertial coefficients [32]. d_p is the pore size of the porous matrix.

2.2 Thermal Model. The porous absorbers are subjected to high fluxes leading to significant variations in local fluid and solid temperatures. Hence, the local thermal non-equilibrium (LTNE) model is applied to obtain respective temperature fields for solid and fluid. Equations (3) and (4) give the energy equations for solid and fluid phases, respectively.

$$(1-\varphi)\rho_s C_{p,s}\frac{\partial T_s}{\partial t}+\nabla\cdot(k_{es}\nabla T_s)+h_v(T_f-T_s)+S_r=0 \quad (3)$$

$$\varphi\left(\rho_f C_{pf}\frac{\partial T_f}{\partial t}\right)+\rho_f C_{pf}\boldsymbol{U}\cdot\nabla T_f=\nabla\cdot(k_{ef}\nabla T_f)+h_v(T_s-T_f) \quad (4)$$

where $k_{es}=\frac{(1-\phi)}{3}k_s$ and $k_{ef}=\phi k_f$ are the effective thermal conductivity for the solid and fluid phases [33]. The energy equations for two phases are coupled via the volumetric heat transfer coefficient h_v. Several correlations are available in the literature for h_v [13,34,35]. The correlation obtained by Wu et al. [36] has been widely used in the literature [16,37–40]. Equation (5) presents the correlation expressed in terms of porosity, pore size, fluid thermal conductivity, k_f, and pore Reynolds number, $\mathrm{Re_p}=\frac{\rho_f U d_p}{\mu}$.

$$\frac{h_v d_p^2}{k_f}=(32.504\phi^{0.38}-109.94\phi^{1.38}+166.65\phi^{2.38}-86.98\phi^{3.38})\mathrm{Re}_p^{0.438} \quad (5)$$

The solid porous matrix is assumed to be gray and optically thick media with absorption and emission. S_r in Eq. (3) is the radiative source term which includes the effect of radiation heat transfer in the solid phase energy equation. The volumetric absorption of incident solar radiation, Q is determined by Beer's Law, given by Eq. (6), where $q(t)$ represents the time varying incident solar flux and $\beta=\frac{3(1-\phi)}{d_p}$ is the extinction coefficient of the porous absorber.

$$Q=q(t)\beta e^{-\beta z} \quad (6)$$

In order to account for energy source due to thermal radiation within the solid porous matrix, the Rosseland approximation [41] is used in the present study. Such an approximation is valid for an optically thick medium where local radiation emitted by the porous matrix travels a very small distance before being attenuated [42]. This assumption allows the radiative source term S_r in Eq. (3) to be transformed into a diffusion equation. For a gray media with constant refractive index, the radiative heat flux can be evaluated by Eq. (7). Rosseland approximation is used by several authors to model radiation heat transfer in porous media [18,43,44].

$$G=-\frac{4\sigma}{3\beta}\nabla(T^4)=k_r\nabla T=-\frac{16\sigma T^3}{3\beta}\nabla T \tag{7}$$

The Rosseland radiation model is implemented by adding the non-linear "radiation thermal conductivity" k_r, to the solid effective thermal conductivity, k_{es}.

2.3 Thermophysical Properties and Boundary Conditions.

The air is treated as a non-participating ideal gas. The temperature-dependent thermophysical properties are used for both air and SiC. The specific heat, C_{pf}, and thermal conductivity of air, k_f, are obtained by polynomial curve fitting to the data set for 100–1600 K [45]. The viscosity is computed by Sutherland Law.

$$C_{pf}=1.93\times10^{-10}T_f^4-8\times10^{-7}T_f^3+1.14\times10^{-3}T_f^2 - 4.49\times10^{-1}T_f+1.06\times10^3 \tag{8}$$

$$k_f=1.52\times10^{-11}T_f^3-4.86\times10^{-8}T_f^2+1.02\times10^{-4}T_f-3.93\times10^{-3} \tag{9}$$

The density of SiC, ρ_s, is taken as 3200 kg/m^3 [46]. Temperature-dependent specific heat capacity, $C_{p,s}$, and thermal conductivity, k_s, of the SiC are given in Eqs. (10) and (11), respectively [47].

$$C_{p,s}=1110+0.15T_s-425e^{-0.0003T_s} \tag{10}$$

$$k_s=\frac{52000e^{-1.24\times10^{-5}T_s}}{T_s+437} \tag{11}$$

Adiabatic and no-slip boundary conditions are applied to the walls. A zero-gauge pressure is applied at the outlet. The radiation heat loss from the solid phase to the ambient is considered at the inlet (Eq. (12). ε is the emissivity of the solid matrix and T_{amb} is the ambient temperature equal to 300 K. σ is the Stefan–Boltzmann constant.

$$k_{es}\nabla T_s=\varepsilon\sigma(T_s^4-T_{amb}^4) \tag{12}$$

3 Validation Studies

The governing equations are solved using Multiphysics Software tool COMSOL 6.0. The LTNE model is validated by comparing the results of the present model with that of Khashan et al. [48]. The domain consisted of a tube filled with porous medium and subjected to a constant wall temperature. The finite volume method was applied to solve the non-dimensional governing equations. Figure 2 presents the LTNE contours for a length equal to one diameter of the tube and Reynolds number, $\mathrm{Re}=100$, Peclet number, $\mathrm{Pe}=30$, Darcy number, $\mathrm{Da}=10^{-4}$, Biot – like number, $\mathrm{Bi}=200$, and thermal conductivity ratio, $\frac{k_{ef}}{k_{es}}=10^{-3}$. A good match can be seen in the results obtained by present study with that of Khashan et al. [48].

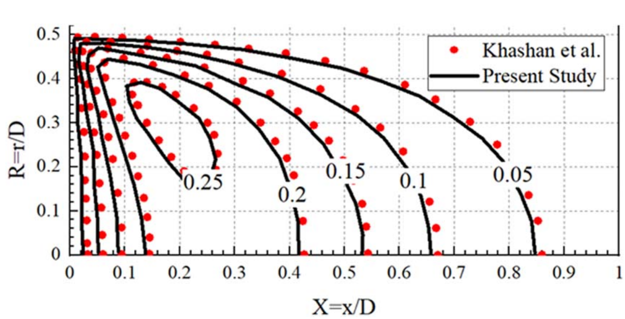

Fig. 2 Comparison of results of Khashan et al. [48] with the present model $\left(\mathrm{Re}=100,\ \mathrm{Pe}=30,\ \mathrm{Da}=10^{-4},\ \mathrm{Bi}=200,\ \frac{k_{ef}}{k_{es}}=10^{-3}\right)$

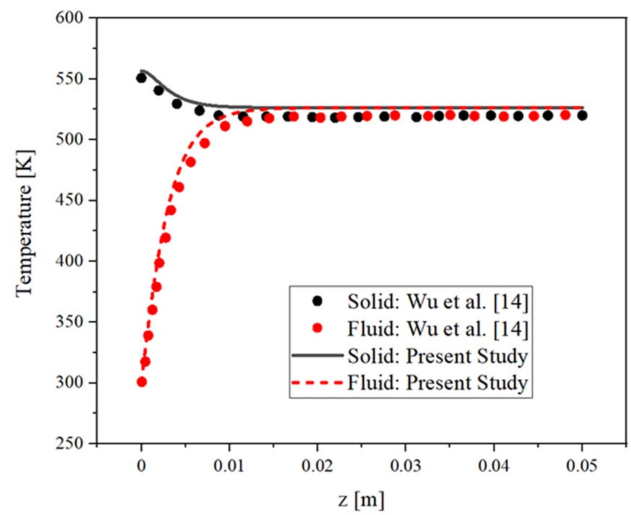

Fig. 3 Comparison of fluid and solid temperature distributions obtained in the present study and Wu et al. [14] ($\phi=0.8$, $d_p=1.5$ mm, $U_{in}=2.16$ m/s)

The validity of the radiation model and the Multiphysics coupling is verified by comparing the results of the present study with that of Wu et al. [14]. The solid and fluid temperature distributions at the centerline for inlet velocity of 2.16 m/s obtained from the present study are compared with that of Wu et al. [14] in Fig. 3.

It is important to note here that the radiation in the present study is modeled by Rosseland approximation, while P1 approximation was used by Wu et al. to solve the radiative heat transfer inside porous media. Additionally, Wu et al. validated their model with experimental results. Hence, the good agreement of the compared results shows that the Rosseland approximation is suitable for modeling radiation transport in porous media under the optically thick assumption.

4 Results and Discussions

The default values of porosity of 0.9, pore size of 2 mm, and inlet velocity of 1 m/s are considered in the present work unless mentioned otherwise. The various scenarios arising during receiver working such as start-up, shut-down, cloud pass, and clear sky operation are implemented in the coupled model by time-dependent incident solar fluxes, which mimic the real working scenarios. It is worth noting that the spatial variations of the flux at the front surface are neglected and only transient fluctuations are considered in the present study. The locations at axial distances of $z=0$ (Point A), $z=L/4$ (Point B), $z=L/2$ (Point C), $z=3L/4$ (Point D), and $z=L$ (Point E) along the centerline are selected to monitor the solid temperature variations within the receiver under different operating conditions. The fluid temperature is not monitored at Point A (inlet) as the temperature at the inlet is assumed to be constant and equal to T_{amb}.

4.1 Cold Start-Up.

The cold start-up of the receiver involves gradual aiming of the incident solar flux from the concentrator onto the receiver. The start-up is implemented in the study by increasing the incident flux from zero to maximum linearly in a period of 60 s and then keeping it constant as given in Eq. (13). A period of 5 s is initially provided to set the flow field. q_{min} and q_{max} are the minimum and maximum incident solar flux and are assumed to

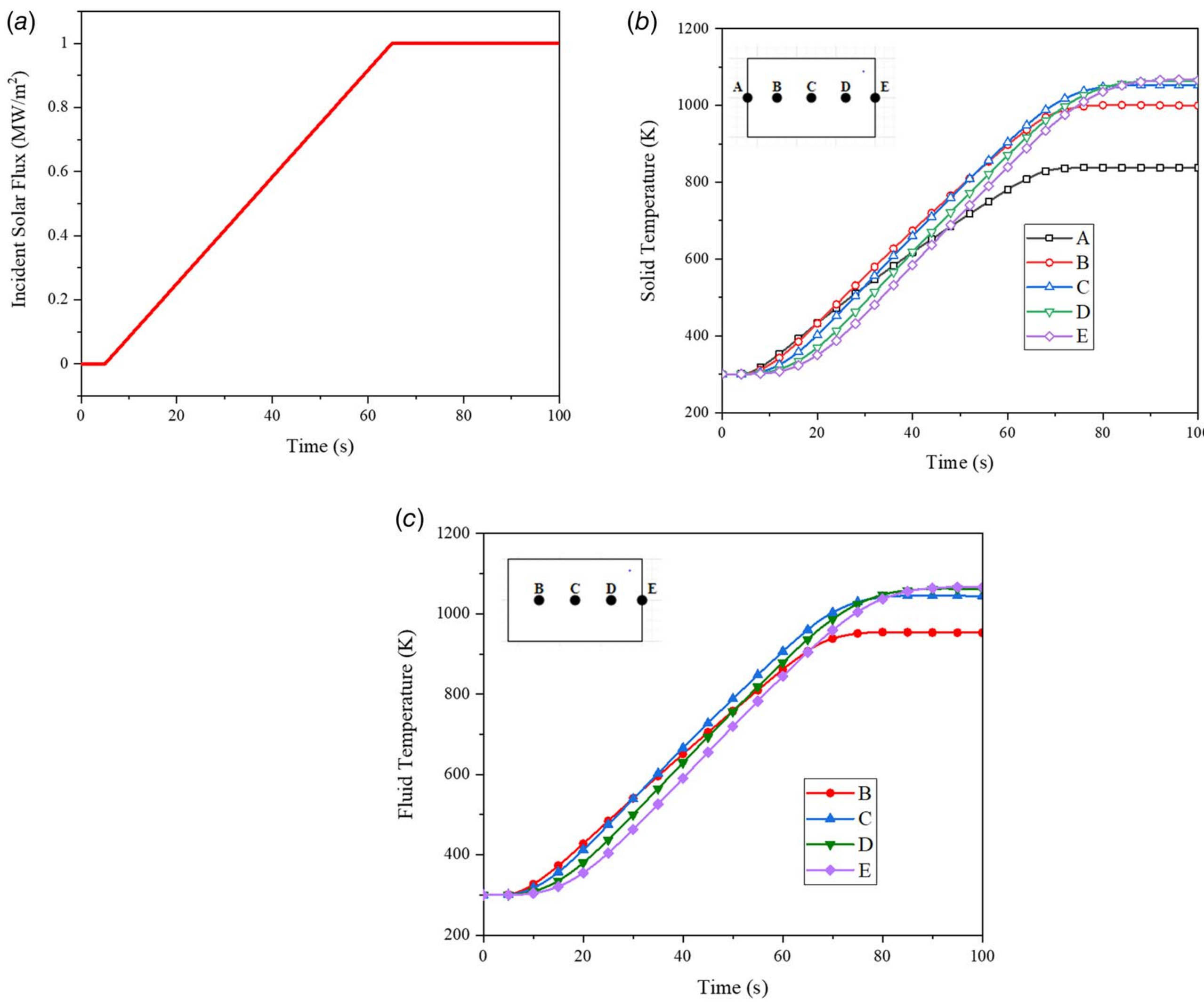

Fig. 4 Transient response of receiver ($\phi = 0.9$, $d_p = 2$ mm, $U_{in} = 1$ m/s) during cold start-up: (*a*) incident solar flux variation, (*b*) solid temperature, and (*c*) fluid temperature

be 0 and 1 $\mathrm{MW/m^2}$, respectively. It equals an increase rate of 16.67 $\mathrm{kW/m^2/s}$ in applied transient solar flux.

$$q(t) = \begin{cases} 0, & t < 5 \\ q_{\min} + \dfrac{(q_{\max} - q_{\min})}{(65-5)}(t-5), & 5 < t \leq 65 \\ q_{\max}, & t \geq 65 \end{cases} \tag{13}$$

The initial temperatures of both solid and fluid phases are initialized at T_{amb} with zero inflow velocity. Figure 4(*a*) presents the solar flux variation during the start-up process. The variation of solid and fluid temperatures at specific locations along the centerline is plotted in Figs. 4(*b*) and 4(*c*), respectively. Point A responds first as it is directly heated by the incident flux. Most of the flux is absorbed within a small length of the receiver only. The other points are heated via conduction and radiation within the solid matrix of the porous receiver. The response time of the locations is directly proportional to their depth from the inlet. The time taken to reach a steady-state follows a similar trend with A reaching the steady-state first. The solid and fluid steady-state temperatures reached by C, D, and E are nearly equal, suggesting that thermal equilibrium is reached beyond C in a steady-state. However, a significant temperature difference exists among different points during start-up, suggesting the possibility of high thermal gradients within the receiver. As the front surface loses heat via radiation directly to the atmosphere, the rate of temperature rise as well as the steady-state temperature of A are significantly less than at other points. The air outlet temperature and pressure drop are plotted in Fig. 5. It can be observed that their variation trend closely follows the incident flux variation. The receiver will start producing a set point outlet temperature of steady-state after about 90 s of the start-up. The solid temperature increases at an average rate of nearly 9.5 K/s when the start-up is implemented in 60 s. However, start-up duration time decides the rate of solar heat flux rise and can significantly affect the rate of solid temperature rise. Figure 6 shows the solid temperature distribution at location C for various start-up durations. For a start-up duration of 0 s, i.e., sudden exposure to maximum solar flux, the absorber reaches steady-state temperature in nearly 30 s. The rate of solid temperature rise is 25 K/s which may result in thermal shocks and can severely affect the structural integrity of the porous matrix.

4.2 Clear Sky Daily Operation.

The change in the sun's position in the sky during the daytime leads to significant variation in the incident solar flux. The time-dependent radiation flux variation obtained using ray-tracing software is used here to approximate the variation on a clear sky day as in Eq. (14) and Fig. 7(*a*).

$$q(t) = q_{av} - q_a \cos(\omega t) \tag{14}$$

where daily average, $q_{av} = 0.7$ $\mathrm{MW/m^2}$, amplitude of variations, $q_a = 0.3$ $\mathrm{MW/m^2}$, and angular frequency, $\omega = \dfrac{2\pi}{8 \times 3600}$ $\mathrm{s^{-1}}$ [28]. A total duration of 8 h is taken to implement the clear sky operation.

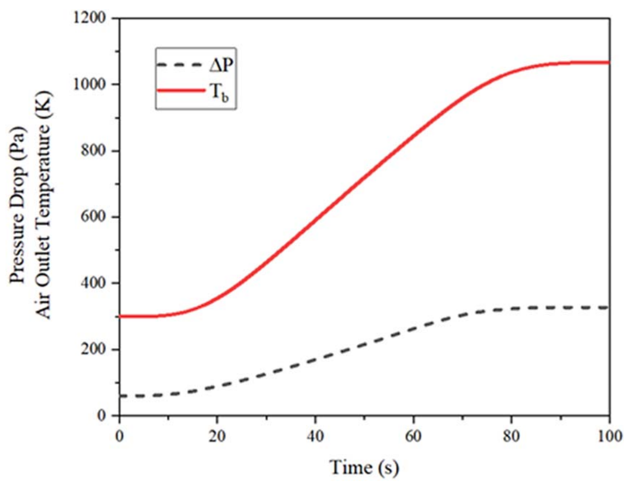

Fig. 5 Air outlet temperature and pressure drop during cold start-up

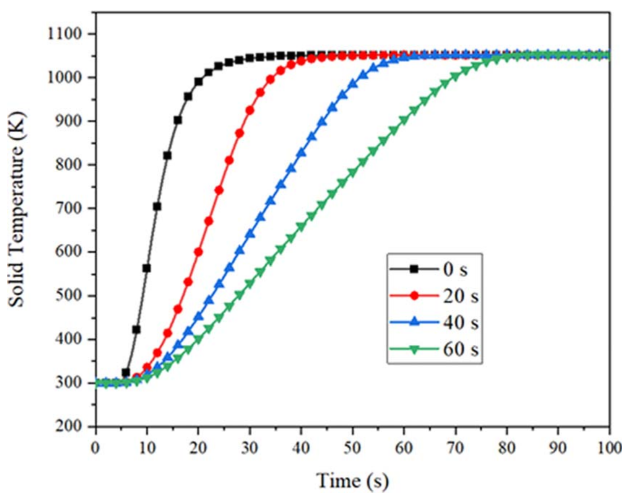

Fig. 6 Solid temperature variation at Point C for different start-up durations

The variation of solid and fluid temperatures at different locations is presented in Figs. 7(*b*) and 7(*c*). It can be observed that solid and fluid temperatures follow the incident flux variation. A significant variation in solid and fluid temperatures occurs during the daily operation. Additionally, the variation of solid temperatures within the receiver indicates non-uniformity in temperature distribution which may induce thermal stresses. Figure 8 presents the air outlet temperature and corresponding pressure drop in the receiver during the daily operation. It can be noted that the air outlet temperature varies up to 450 K during the daily operation. Such a high variation during working hours might affect the downstream processes. It also indicates a smooth pressure drop variation from 65 Pa to 326.8 Pa during the entire working hours with a maximum during mid-hours when the incident flux is maximum.

4.3 Cloud Passing. The cloud passing phenomena may result in partial or complete loss of incident solar flux. Figure 9(*a*) shows a typical case of cloud passage which includes covering of concentrator, fully covered duration, and subsequent uncovering of the concentrator [49]. The covering duration is implemented by reducing incident flux from maximum to zero in 5 s. The reduction of flux to zero represents the worst case. The flux is kept at zero for the next 30 s, after which uncovering of the cloud takes place. The uncovering duration is implemented with an increase of incident flux on the receiver to a maximum of 5 s after which the flux remains constant at maximum value. The whole variation is expressed in Eq. (15). $q_{\min}$ and $q_{\max}$ are the minimum and maximum incident solar flux and are assumed to be 0 and 1 MW/m^2, respectively. A rate of change of flux of 0.2 MW/m^2/s occurs during the covering and uncovering time.

$$q(t) = \begin{cases} q_{\max}, & 0 < t \le 5 \\ q_{\max} - \dfrac{(q_{\max} - q_{\min})}{(10-5)}(t-5), & 5 < t \le 10 \\ q_{\min}, & 10 < t \le 40 \\ q_{\min} + \dfrac{(q_{\max} - q_{\min})}{(45-40)}(t-40), & 40 < t \le 45 \\ q_{\max}, & t > 45 \end{cases} \tag{15}$$

In the first stage of simulation, the receiver is allowed to reach a steady-state. The steady-state values are then used in the second transient stage of the simulation as initial conditions. A similar approach was used by Wu et al. [24] to perform transient simulations during sudden loss of heat flux. The solid and fluid temperatures at the different locations are presented in Figs. 9(*b*) and 9(*c*), respectively. It can be observed that solid and fluid temperatures at all points reach close to the ambient temperature during the cloud pass. Since the interior locations mostly get heated indirectly via conduction and radiation and not by direct incident flux, their thermal response is delayed compared to Point A at the front. At any given instant, the solid temperature distribution within the receiver is extremely non-uniform. This might lead to very large thermal gradients. Additionally, the solid temperature falls from a steady-state to ambient and again rises to a steady-state within a short duration of 50 s which may result in thermal shock and failure of porous material. This observation suggests that frequent cloud passing may result in severe thermal cycles and fatigue failure of the receiver. The air outlet temperature and pressure drop in the receiver are presented in Fig. 10. The receiver will start producing air temperature lower than 600 K within 25 s and reach close to ambient temperature in about 40 s. The time required by the receiver to again reach the steady-state after the cloud pass is about 45 s. The sudden variation in air outlet temperature might disrupt the further operations of the plant. The study so far in this section considered the worst case cloud type with zero transmissivity. Cloud transmissivity determines the ratio of sunlight passing through it [49]. To understand the absorber behavior under different types of clouds, the solid temperature variation at location C for different cloud transmissivities is presented in Fig. 11. The various transmissivities are implemented in the model by varying the minimum flux, $q_{\min}$. It can be clearly observed that for the same duration, the variation in solid temperatures is much higher for clouds with lower transmissivities and vice-versa.

4.4 Shut-Down. The gradual shut-down of the plant is implemented in the present study. Similar to the start-up period, the incident flux is linearly reduced from steady-state to a minimum value in 60 s. This variation is expressed by Eq. (16) and is shown in Fig. 12(*a*). $q_{\min}$ and $q_{\max}$ are the minimum and maximum incident solar fluxes and are assumed to be 0 and 1 MW/m^2, respectively.

$$q(t) = \begin{cases} q_{\max}, & t < 5 \\ q_{\max} - \dfrac{(q_{\max} - q_{\min})}{(65-5)}(t-5), & 5 < t \le 65 \\ q_{\min}, & t \ge 65 \end{cases} \tag{16}$$

The same two-stage strategy used in Sec. 4.3 is implemented to obtain initial steady-state conditions and transient solutions for the receiver. The corresponding solid and fluid temperature variations are plotted in Figs. 12(*b*) and 12(*c*), respectively. As observed in earlier cases, Point A responds first, followed by other locations depending on their respective locations from the front. The farthest

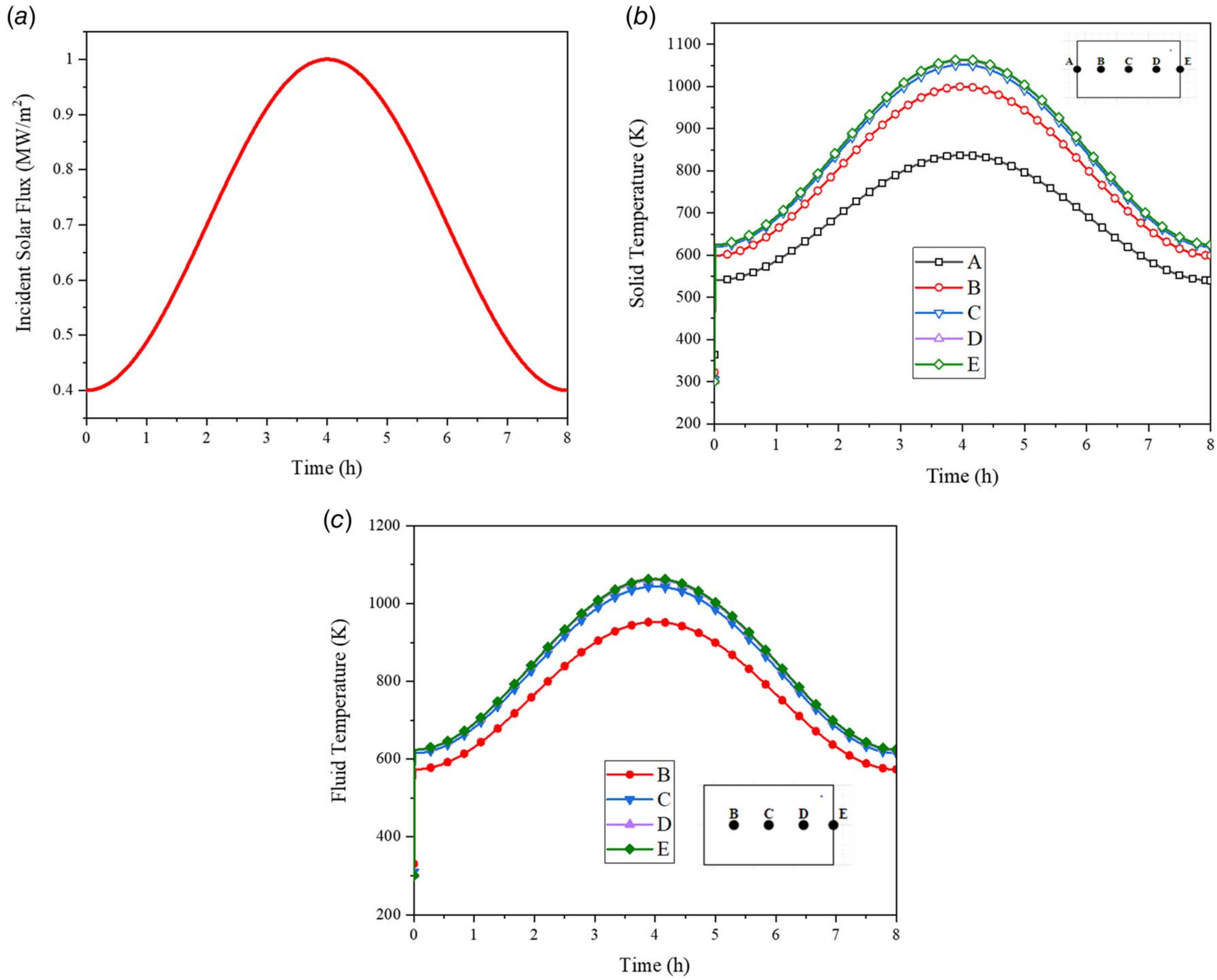

Fig. 7 Transient response of receiver ($\phi = 0.9$, $d_p = 2$ mm, $U_{in} = 1$ m/s) during clear sky operation: (*a*) incident solar flux variation, (*b*) solid temperature, and (*c*) fluid temperature

point, E, reaches the ambient conditions last, nearly 30 s after the incident flux is cut down to zero. A similar trend is observed in fluid temperatures as well. Figure 13 presents the air outlet temperature and pressure drop in the receiver during the shut-down period.

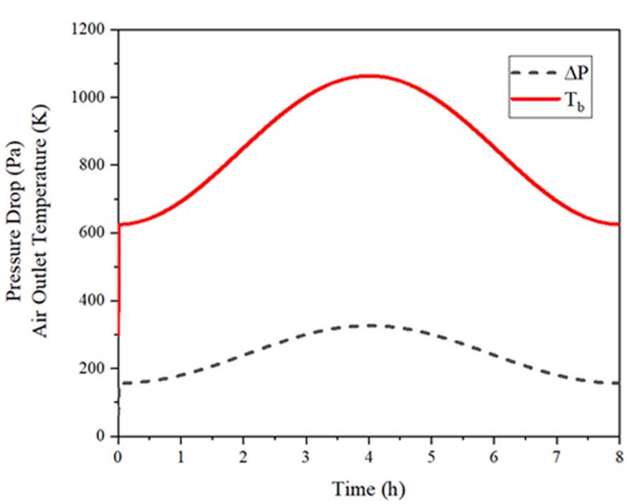

Fig. 8 Pressure drop and air outlet temperature during clear sky operation

It can be seen that the receiver takes about 80 s to reach the ambient temperature. The average rate of change in solid temperature is about 10 K/s. In order to investigate the effect of the shut-down period, Fig. 14 presents the solid temperature variation for various shut-down periods. It can be observed that a shut-down duration of 0 s, i.e., a sudden drop of incident flux from the maximum value to zero, results in a solid temperature change rate of nearly 38 K/s which is extremely high and may cause thermal shock in the porous matrix. As the shut-down period increases, the rate of solid temperature change reduces, decreasing the possibility of absorber failure.

4.5 Effect of Structural Parameters.

The porosity and pore size are two major structural parameters that determine the fluid flow and heat transfer and radiative properties of a porous medium. The variation in porosity and pore size significantly affects the flow field, pressure drop, and solid and fluid temperature distributions. Hence, it is important to understand their effect on the transient response of the receiver under various conditions. This section investigates the effect of porosity and pore size on the solid temperature distribution. The porosity values of 0.7, 0.8, and 0.9 are taken to investigate the effect of porosity. Similarly, pore sizes of 2, 3, and 4 mm are taken to investigate the effect of pore size. The solid temperature at $z = L/2$ (Point C) is plotted in cases of cold start-up, cloud pass, and shut-down.

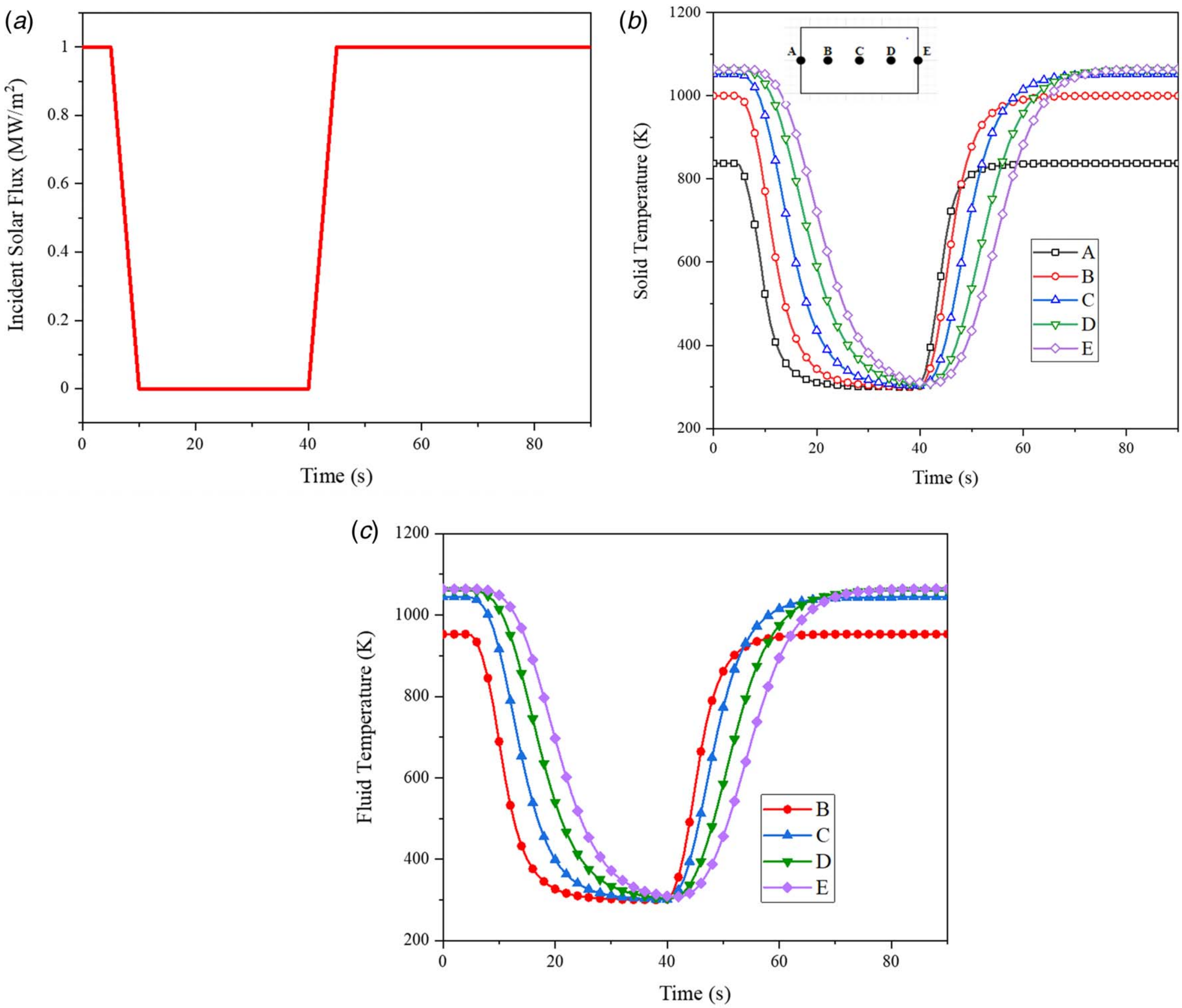

Fig. 9 Transient response of receiver ($\phi = 0.9$, $d_p = 2$ mm, $U_{in} = 1$ m/s) during cloud passing: (*a*) incident solar flux variation, (*b*) solid temperature, and (*c*) fluid temperature

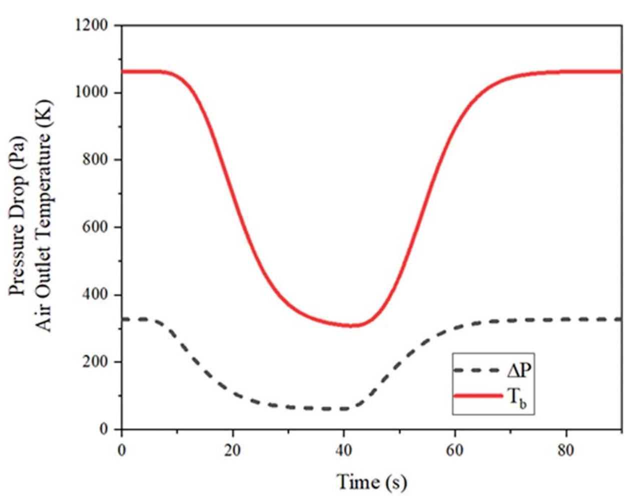

Fig. 10 Pressure drop and air outlet temperature during cloud passing

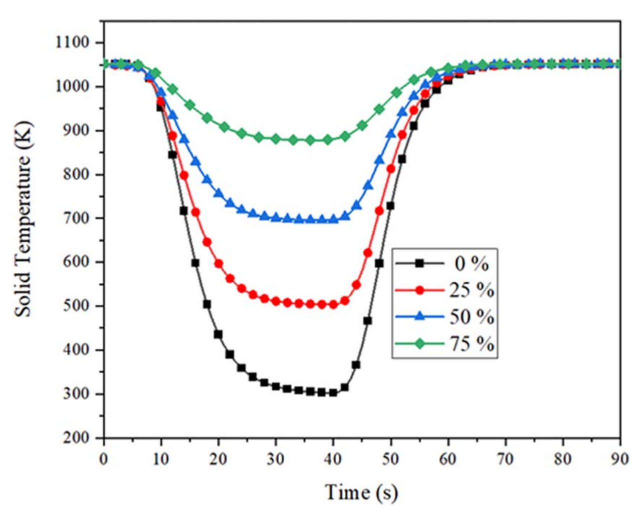

Fig. 11 Solid temperature variation at location C for various cloud transmissivities

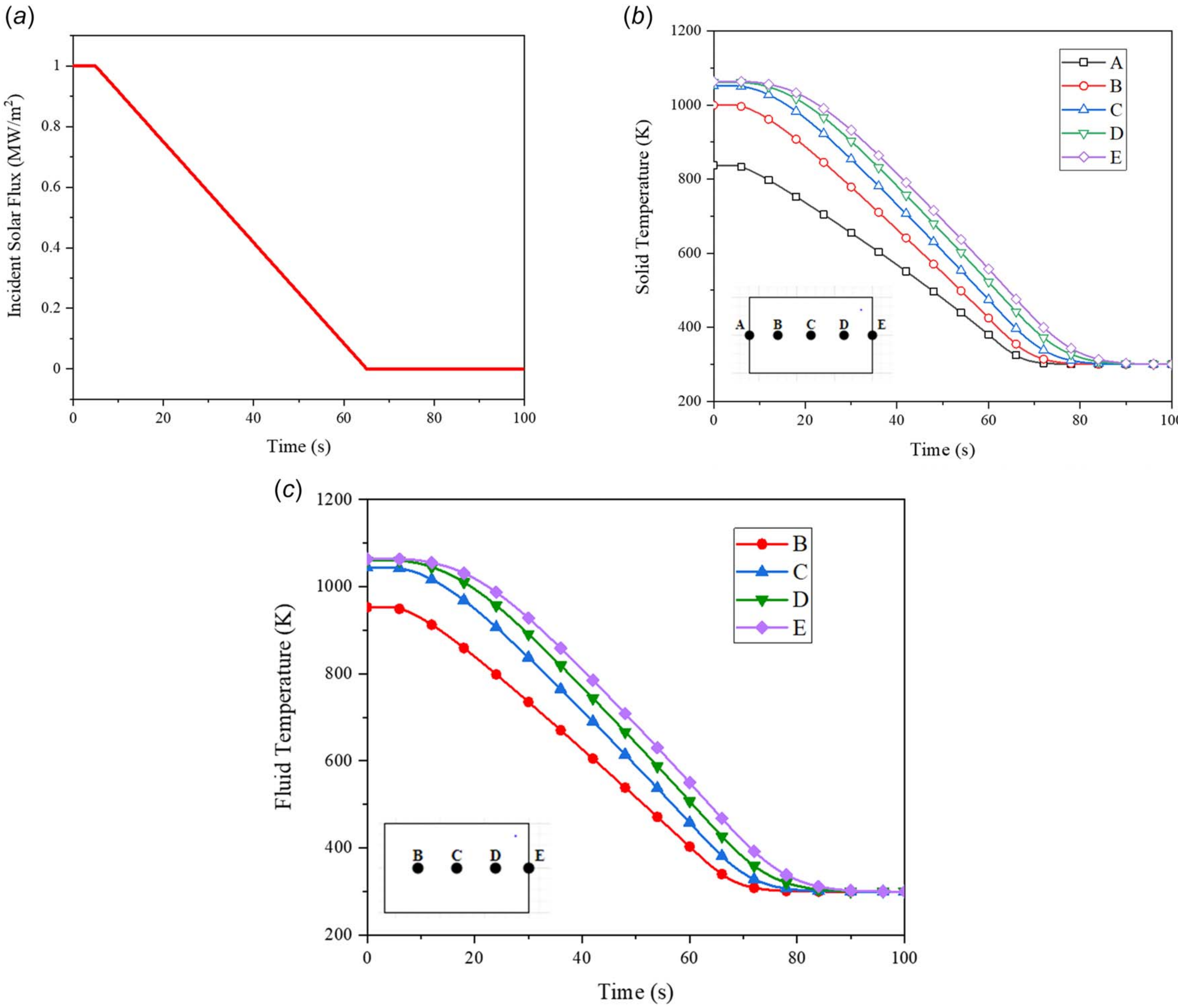

Fig. 12 Transient response of receiver ($\phi = 0.9$, $d_p = 2$ mm, $U_{in} = 1$ m/s) during shut-down: (*a*) incident solar flux distribution, (*b*) solid temperature, and (*c*) fluid temperature

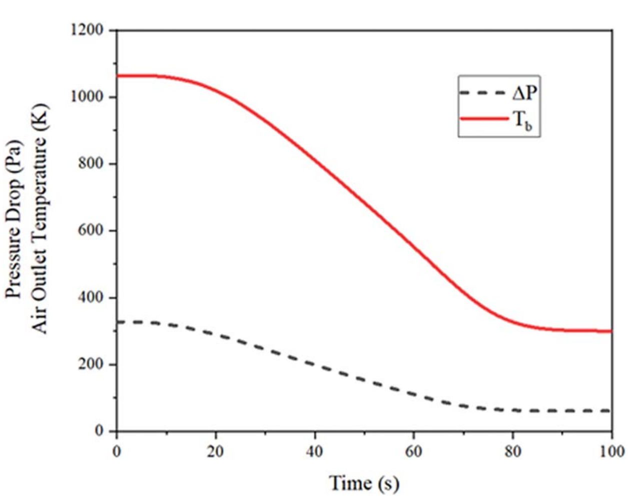

Fig. 13 Air outlet temperature and pressure drop during shut-down period

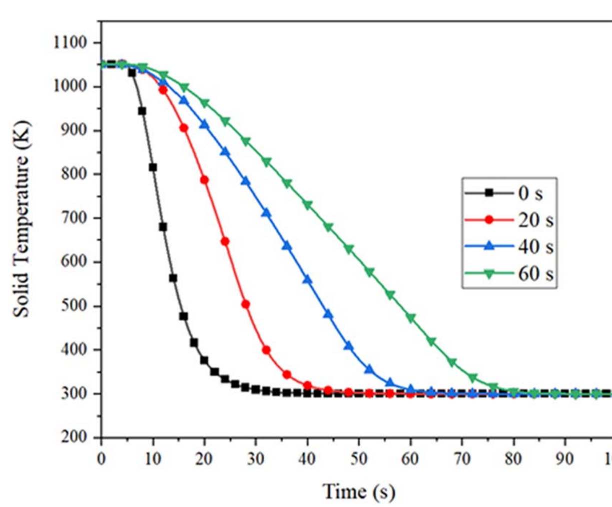

Fig. 14 Solid temperature variation in receiver ($\phi = 0.9$, $d_p = 2$ mm, $U_{in} = 1$ m/s) for different shut-down periods

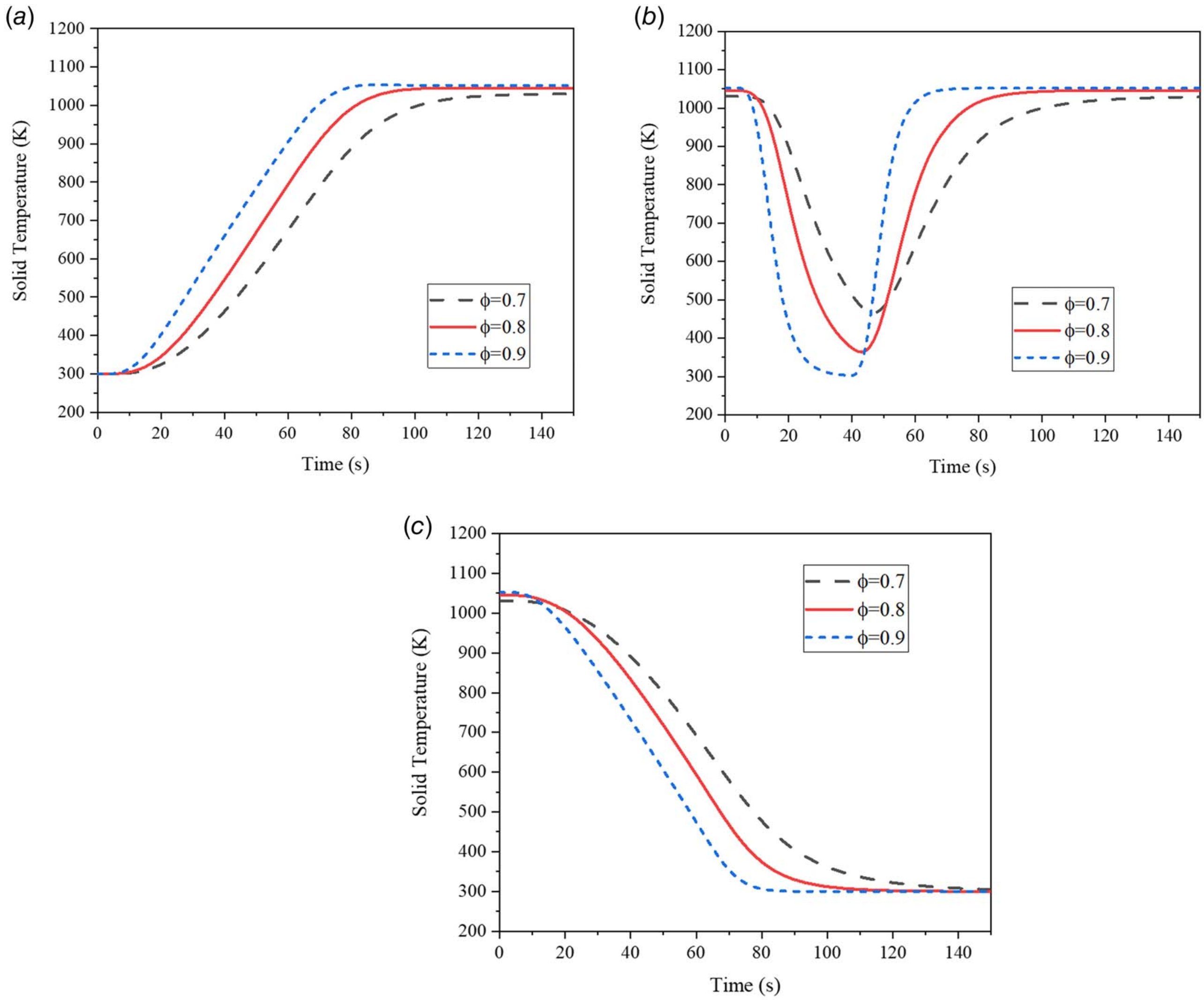

Fig. 15 Effect of porosity on solid temperature distribution of receiver ($d_p = 2$ mm, $U_{in} = 1$ m/s) at $z = L/2$ (Point C): (*a*) cold start-up, (*b*) cloud pass, and (c) shut-down

4.5.1 Effect of Porosity. The solid temperature variation with time under different operations for selected porosities is presented in Fig. 15. It can be clearly seen that the porosity of 0.9 shows a faster transient response to change in incident fluxes in all cases, followed by 0.8 and 0.7, respectively. The rate of temperature change is greater for higher porosities, as evident from the steep rise and fall of temperatures with flux variations in Figs. 15(*a*) and 15(*c*). Conversely, smaller porosities are slower to respond to flux changes. A similar observation was made by [26]. Thus, the thermal shock is more probable in the case of higher porosities due to a greater temperature change rate. It can also be noted that temperature change becomes more linear as the porosity increases from 0.7 to 0.9. As discussed in Sec. 4.3, frequent cloud passes may lead to severe thermal cycles. Hence, the choice of material should have a slow thermal response, ensuring that the temperature fluctuation rates are within limits. Figure 15(*b*) suggests that temperature fluctuation is highest for the porosity of 0.9 porosity while least for that of 0.7 during a cloud pass. The solid temperature drops to ambient temperature within 40 s for the porosity of 0.9 while the drop in the case of 0.7 is only up to 467 K in about 50 s. However, the faster response is beneficial in quickly restoring the original steady-state working conditions when dynamic control is unavailable. The time taken to reach a steady-state in case of cold start-up and shut-down is about 90 s, 110 s, and 130 s for the porosity of 0.9, 0.8, and 0.7, respectively. Similarly, steady-state is restored in 70 s with porosity of 0.9 during cloud pass while 90 s and 120 s are required in case of porosities of 0.8 and 0.7, respectively. Since fluid temperature variation follows the same trend as that of solid as observed in previous sections, the air outlet temperature will quickly reach the steady-state point. This will restore the output air temperature as it was prior to flux fluctuations in a shorter period.

4.5.2 Effect of Pore Size. The solid temperature variations under different operating conditions are plotted in Fig. 16. It can be clearly observed that the effect of pore size is lesser as compared to porosity. In addition, a slightly steeper rise and fall of temperature are noted for a pore diameter of 2 mm in all cases. This suggests that smaller pore sizes have a faster thermal response. The temperature rise is linear for the start-up and shut-down period at the constant porosity of 0.9, as observed in Figs. 16(*a*) and 16(*c*). This suggests that the shape of the time-dependent temperature curve is controlled mostly by porosity. The time to reach the steady-state is also nearly the same for all the pore sizes considered in the study at a constant porosity. During cloud pass, the temperature drop is slightly less for the pore size of 4 mm as seen in Fig. 16(*b*). The temperature dips down till 40 s, after which it rises till the steady-state prior to cloud pass is restored at about 90 s. Since lower pore sizes enhance convective heat transfer, the steady-state temperature reached for a pore size of 2 mm is the highest during start-up and cloud pass. The nearly overlapping curves in Fig. 16(*c*) indicate

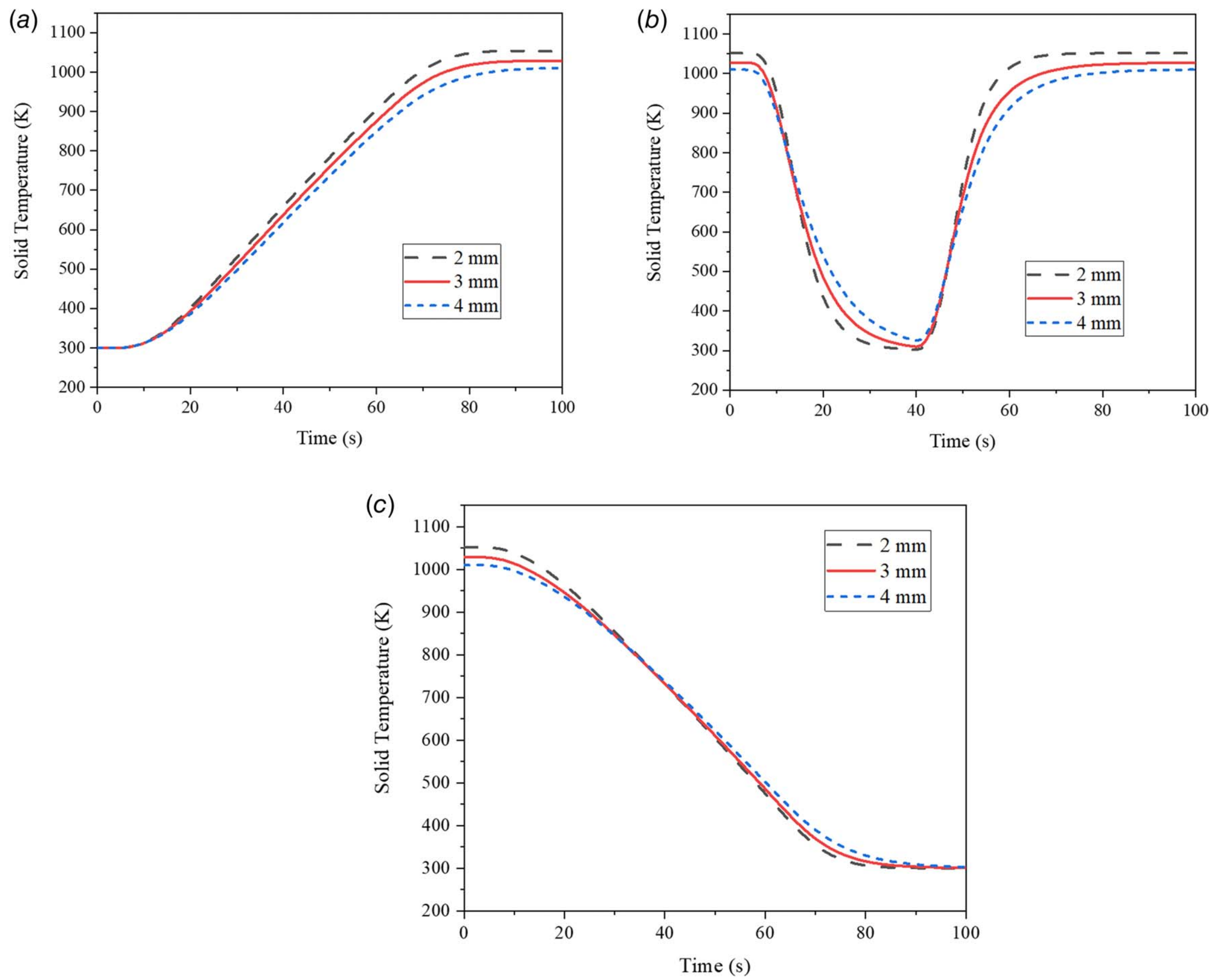

Fig. 16 Effect of pore size on solid temperature distribution of receiver ($\phi = 0.9$, $U_{in} = 1$ m/s) at location $z = L/2$ (Point C): (*a*) cold start-up, (*b*) cloud pass, and (c) shut-down

that the rate of drop in temperature remains nearly the same for all pore sizes during the shut-down period.

5 Conclusions

The present paper extends the understanding of the transient behavior of a porous volumetric receiver under different operating conditions. A coupled thermal transient model is developed, and the dynamic responses of the receiver during start-up, shut-down, cloud pass, and clear sky conditions are investigated. The effect of porosity and pore size on the transient working of the receiver is also studied in detail. It is observed that the fluid and solid temperature variations at different locations, air outlet temperature, and pressure drop variation closely follow the same trend as that of incident solar heat flux variation. The time taken by various locations within the receiver to respond to flux changes increases with depth. Consequently, the time taken to reach steady-state temperatures follows the same trend with the front reaching the steady-state first. Significant thermal gradients exist within the receiver during transient flux periods. The temperature change rate depends on the rate of increase or decrease in incident flux. Smaller start-up and shut-down periods result in higher temperature variation and vice-versa. The effect of the cloud pass phenomena on the receiver's dynamic performance depends on the clouds' transmissivity, with zero transmissivity clouds leading to the worst gradients. Such frequent cloud passes may induce thermal fatigue in the porous receiver structure. In addition, the outlet air temperature varies up to 450 K during a clear sky operation which can adversely affect the downstream processes. It indicates the requirement of robust dynamic control for the receivers. Regarding the influence of porous structural parameters, it is observed that the effect of porosity on the transient working of the receiver is more significant as compared to pore size. The higher porosity results in a steeper rise and fall of temperature in the receiver. The rate of change of temperature for different pore sizes remains nearly the same under all conditions. The results of the study provide major conclusions which can be used to predict the transient thermal behavior for better design of receiver operations. The knowledge of thermal gradients can be used to predict the thermal stresses induced within the receiver and to ensure its structural integrity. Additionally, the results may help in monitoring and dynamic control of various parameters during its operation.

Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

k = thermal conductivity (W/mK)
p = pressure (Pa)
q = incident solar radiation (W/m^2)
D = diameter (m)
G = radiative heat flux (W/m^2)
K = permeability (m^2)
L = length (m)
Q = volumetric heat source (W/m^3)
T = temperature (K)
$\boldsymbol{U}$ = Darcy velocity (m/s)
h_v = volumetric heat transfer coefficient (W/m^{2}K)
d_p = pore size (m)
C_F = inertial coefficient
C_p = specific heat (J/kgK)
S_r = radiative source term (W/m^3)
r, z = radial and axial coordinates (m)

Greek Symbols

β = extinction coefficient (m^{-1})
ε = emissivity
μ = dynamic viscosity (Pa $\cdot$ s)
ρ = density (kg/m^3)
σ = Stefan–Boltzmann constant (W/m^{2}K^4)
ϕ = porosity
ω = angular frequency (1/s)

Subscripts

c = collimated
d = diffuse
e = effective
f = fluid
r = radiation
s = solid
w = wall
av = average
in = inlet
amb = ambient

Abbreviations

FVM = finite volume method
LTNE = local thermal non-equilibrium
MCRT = Monte Carlo ray tracing
SiC = silicon carbide

References

[1] Lund, H., 2007, "Renewable Energy Strategies for Sustainable Development," Energy, **32**(6), pp. 912–919.

[2] Kannan, N., and Vakeesan, D., 2016, "Solar Energy for Future World:—A Review," Renewable Sustainable Energy Rev., **62**, pp. 1092–1105.

[3] He, Y.-L., Qiu, Y., Wang, K., Yuan, F., Wang, W.-Q., Li, M.-J., and Guo, J.-Q., 2020, "Perspective of Concentrating Solar Power," Energy, **198**, p. 117373.

[4] Majidi, M., Behbahaninia, A., Amidpour, M., and Sadati, S. H., 2021, "Thermoeconomic Optimization of a Novel High-Efficiency Combined-Cycle Hybridization With a Solar Power Tower System," Energy Convers. Manage., **244**, p. 114461.

[5] Luo, Y., Lu, T., and Du, X., 2018, "Novel Optimization Design Strategy for Solar Power Tower Plants," Energy Convers. Manage., **177**, pp. 682–692.

[6] Ho, C. K., and Iverson, B. D., 2014, "Review of High-Temperature Central Receiver Designs for Concentrating Solar Power," Renewable Sustainable Energy Rev., **29**(1), pp. 835–846.

[7] He, Y. L., Wang, K., Qiu, Y., Du, B. C., Liang, Q., and Du, S., 2019, "Review of the Solar Flux Distribution in Concentrated Solar Power: Non-Uniform Features, Challenges, and Solutions," Appl. Therm. Eng., **149**, pp. 448–474.

[8] Ávila-Marín, A. L., 2011, "Volumetric Receivers in Solar Thermal Power Plants With Central Receiver System Technology: A Review," Sol. Energy, **85**(5), pp. 891–910.

[9] Capuano, R., Fend, T., Schwarzbözl, P., Smirnova, O., Stadler, H., Hoffschmidt, B., and Pitz-Paal, R., 2016, "Numerical Models of Advanced Ceramic Absorbers for Volumetric Solar Receivers," Renewable Sustainable Energy Rev., **58**, pp. 656–665.

[10] Ali, M., Rady, M., Attia, M. A. A., and Ewais, E. M. M., 2020, "Consistent Coupled Optical and Thermal Analysis of Volumetric Solar Receivers With Honeycomb Absorbers," Renewable Energy, **145**, pp. 1849–1861.

[11] Nakakura, M., Bellan, S., Matsubara, K., and Kodama, T., 2018, "Conjugate Radiation-Convection-Conduction Simulation of Volumetric Solar Receivers With Cut-Back Inlets," Sol. Energy, **170**, pp. 606–617.

[12] Nakakura, M., Matsubara, K., Bellan, S., and Kodama, T., 2020, "Direct Simulation of a Volumetric Solar Receiver With Different Cell Sizes at High Outlet Temperatures (1,000–1,500 °C)," Renewable Energy, **146**, pp. 1143–1152.

[13] Xu, C., Song, Z., Chen, L. D., and Zhen, Y., 2011, "Numerical Investigation on Porous Media Heat Transfer in a Solar Tower Receiver," Renewable Energy, **36**(3), pp. 1138–1144.

[14] Wu, Z., Caliot, C., Flamant, G., and Wang, Z., 2011, "Coupled Radiation and Flow Modeling in Ceramic Foam Volumetric Solar Air Receivers," Sol. Energy, **85**(9), pp. 2374–2385.

[15] Barreto, G., Canhoto, P., and Collares-Pereira, M., 2018, "Three-Dimensional Modelling and Analysis of Solar Radiation Absorption in Porous Volumetric Receivers," Appl. Energy, **215**, pp. 602–614.

[16] Barreto, G., Canhoto, P., and Collares-Pereira, M., 2020, "Parametric Analysis and Optimisation of Porous Volumetric Solar Receivers Made of Open-Cell SiC Ceramic Foam," Energy, **200**, p. 117476.

[17] Roldán, M. I., Smirnova, O., Fend, T., Casas, J. L., and Zarza, E., 2014, "Thermal Analysis and Design of a Volumetric Solar Absorber Depending on the Porosity," Renewable Energy, **62**, pp. 116–128.

[18] Du, S., Xia, T., He, Y. L., Li, Z. Y., Li, D., and Xie, X. Q., 2020, "Experiment and Optimization Study on the Radial Graded Porous Volumetric Solar Receiver Matching Non-Uniform Solar Flux Distribution," Appl. Energy, **275**(May), p. 115343.

[19] Laporte-Azcué, M., González-Gómez, P. A., Rodríguez-Sánchez, M. R., and Santana, D., 2022, "A Procedure to Predict Solar Receiver Damage During Transient Conditions," Renewable Sustainable Energy Rev., **154**, p. 111905.

[20] Soo Too, Y. C., García, J., Padilla, R. V., Kim, J. S., and Sanjuan, M., 2019, "A Transient Optical-Thermal Model With Dynamic Matrix Controller for Solar Central Receivers," Appl. Therm. Eng., **154**, pp. 686–698.

[21] Xu, L., Stein, W., Kim, J. S., Too, Y. C. S., Guo, M., and Wang, Z., 2018, "Transient Numerical Model for the Thermal Performance of the Solar Receiver," Appl. Therm. Eng., **141**, pp. 1035–1047.

[22] Ren, Y., Qi, H., Shi, J., Chen, Q., Wang, Y., and Ruan, L., 2017, "Thermal Performance Characteristics of Porous Media Receiver Exposed to Concentrated Solar Radiation," J. Energy Eng., **143**(5), p. 04017013.

[23] Reddy, K. S., and Nataraj, S., 2019, "Thermal Analysis of Porous Volumetric Receivers of Concentrated Solar Dish and Tower Systems," Renewable Energy, **132**, pp. 786–797.

[24] Wu, Z., and Wang, Z., 2013, "Fully Coupled Transient Modeling of Ceramic Foam Volumetric Solar Air Receiver," Sol. Energy, **89**, pp. 122–133.

[25] Fuqiang, W., Zhennan, G., Jianyu, T., Lanxin, M., Zhenyu, Y., and Heping, T., 2016, "Transient Thermal Performance Response Characteristics of Porous-Medium Receiver Heated by Multi-dish Concentrator," Int. Commun. Heat Mass Transfer, **75**, pp. 36–41.

[26] Ribeiro, R. R., and de Lemos, M. J. S., 2020, "Transient Behavior and Thermal Efficiency of Volumetric Heat Receivers," Int. J. Heat Mass Transfer, **149**, p. 119128.

[27] Zoller, S., Koepf, E., Roos, P., and Steinfeld, A., 2019, "Heat Transfer Model of a 50 KW Solar Receiver-Reactor for Thermochemical Redox Cycling Using Cerium Dioxide," ASME J. Sol. Energy Eng., **141**(2), p. 021014.

[28] Herrmann, B., Behzad, M., Cardemil, J. M., Calderón-Muñoz, W. R., and Fernández, R. M., 2020, "Conjugate Heat Transfer Model for Feedback Control and State Estimation in a Volumetric Solar Receiver," Sol. Energy, **198**, pp. 343–354.

[29] Avila-Marin, A. L., Fernandez-Reche, J., and Martinez-Tarifa, A., 2019, "Modelling Strategies for Porous Structures as Solar Receivers in Central Receiver Systems: A Review," Renewable Sustainable Energy Rev., **111**, pp. 15–33.

[30] Casalegno, V., Ferrari, L., Fuentes, M. J., De Zanet, A., Gianella, S., Ferraris, M., and Candelario, V. M., 2021, "High-Performance Sic–Based Solar Receivers for CSP: Component Manufacturing and Joining," Materials (Basel), **14**(16), pp. 1–17.

[31] 2021, "CFD Module User's Guide," COMSOL Multiphysics, COMSOL 6.0, pp. 623–631.

[32] Wu, Z., Caliot, C., Bai, F., Flamant, G., Wang, Z., Zhang, J., and Tian, C., 2010, "Experimental and Numerical Studies of the Pressure Drop in Ceramic Foams for Volumetric Solar Receiver Applications," Appl. Energy, **87**(2), pp. 504–513.

[33] Schuetz, M. A., and Glicksman, L. R., 1984, "A Basic Study of Heat Transfer Through Foam Insulation," J. Cell. Plast., **20**(2), pp. 114–121.

[34] Du, S., Tong, Z. X., Zhang, H. H., and He, Y. L., 2019, "Tomography-Based Determination of Nusselt Number Correlation for the Porous Volumetric Solar Receiver With Different Geometrical Parameters," Renewable Energy, **135**, pp. 711–718.

[35] Wang, P., Li, J. B., Xu, R. N., and Jiang, P. X., 2021, "Non-Uniform and Volumetric Effect on the Hydrodynamic and Thermal Characteristic in a Unit Solar Absorber," Energy, **225**, p. 120130.

[36] Wu, Z., Caliot, C., Flamant, G., and Wang, Z., 2011, "Numerical Simulation of Convective Heat Transfer Between Air Flow and Ceramic Foams to Optimise Volumetric Solar Air Receiver Performances," Int. J. Heat Mass Transfer, **54**(7–8), pp. 1527–1537.

[37] Li, X. L., Sun, C., Xia, X. L., Li, Z. H., and Li, Y., 2020, "Modeling of Coupled Heat Transfer in a Windowed Volumetric Solar Receiver," Sol. Energy, **201**, pp. 195–208.

[38] Chen, X., Xia, X. L., Yan, X. W., and Sun, C., 2017, "Heat Transfer Analysis of a Volumetric Solar Receiver With Composite Porous Structure," Energy Convers. Manage., **136**, pp. 262–269.

[39] Chen, X., Xia, X. L., Meng, X. L., and Dong, X. H., 2015, "Thermal Performance Analysis on a Volumetric Solar Receiver With Double-Layer Ceramic Foam," Energy Convers. Manage., **97**, pp. 282–289.

[40] Nimvari, M. E., Jouybari, N. F., and Esmaili, Q., 2018, "A New Approach to Mitigate Intense Temperature Gradients in Ceramic Foam Solar Receivers," Renewable Energy, **122**, pp. 206–215.

[41] Howell, J. R., Menguc, M. P., and Siegel, R., 2016, *Thermal Radiation Heat Transfer*, CRC Press, Boca Raton, FL, pp. 576–578.

[42] Gomez-Garcia, F., González-Aguilar, J., Olalde, G., and Romero, M., 2016, "Thermal and Hydrodynamic Behavior of Ceramic Volumetric Absorbers for Central Receiver Solar Power Plants: A Review," Renewable Sustainable Energy Rev., **57**, pp. 648–658.

[43] Wang, F., Shuai, Y., Tan, H., and Yu, C., 2013, "Thermal Performance Analysis of Porous Media Receiver With Concentrated Solar Irradiation," Int. J. Heat Mass Transfer, **62**, pp. 247–254.

[44] Sano, Y., Iwase, S., and Nakayama, A., 2012, "A Local Thermal Nonequilibrium Analysis of Silicon Carbide Ceramic Foam as a Solar Volumetric Receiver," ASME J. Sol. Energy Eng., **134**(2), p. 021006.

[45] Hurley, M. J., 2016, *SFPE Handbook of Fire Protection Engineering*, Springer, New York, p. 3426.

[46] Chen, X., Xia, X. L., Sun, C., and Yan, X. W., 2017, "Transient Thermal Analysis of the Coupled Radiative and Convective Heat Transfer in a Porous Filled Tube Exchanger at High Temperatures," Int. J. Heat Mass Transfer, **108**, pp. 2472–2480.

[47] Munro, R. G., 1997, "Material Properties of a Sintered α-SiC," J. Phys. Chem. Ref. Data, **26**(5), pp. 1195–1203.

[48] Khashan, S. A., Al-Amiri, A. M., and Al-Nimr, M. A., 2005, "Assessment of the Local Thermal Non-Equilibrium Condition in Developing Forced Convection Flows Through Fluid-Saturated Porous Tubes," Appl. Therm. Eng., **25**(10), pp. 1429–1445.

[49] Augsburger, G., and Favrat, D., 2013, "Modelling of the Receiver Transient Flux Distribution Due to Cloud Passages on a Solar Tower Thermal Power Plant," Sol. Energy, **87**, pp. 42–52.